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Кафедра ИИТ

Лабораторная работа №3
По дисциплине «Математические основы интеллектуальных систем»
За 3 семестр
Тема: «Нахождение кратчайшего пути в графе»

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Лабораторная работа №3

«Нахождение кратчайшего пути в графе»

Цель работы: освоить алгоритмы Дейкстры и Флойда–Уоршелла для нахождения кратчайших путей в взвешенном графе и реализовать их на языке C.

Ход работы

Код

```
#define _CRT_SECURE_NO_WARNINGS
#include <stdio.h>
#include <stdlib.h>
// "Бесконечность" — большое число, влезает в long long при сложении
#define INF 10000000000000000LL
// ПРЕДСТАВЛЕНИЕ ГРАФА: СПИСКИ СМЕЖНОСТИ (со весами)
typedef struct Edge {
    int v;              // соседняя вершина
    long long w;        // вес ребра
    struct Edge* next;  // следующий сосед в списке
} Edge;
typedef struct {
    int   n;            // число вершин
    int   m;            // число рёбер
    Edge** adj;         // adj[i] — голова списка соседей вершины i
} Graph;
// Добавление неориентированного взвешенного ребра
static void add_edge(Graph* g, int u, int v, long long w)
{
    // Добавляем ребро в список u: сосед — v
    Edge* p = (Edge*)malloc(sizeof(Edge));
    p->v = v; p->w = w; p->next = g->adj[u]; g->adj[u] = p;
    // Добавляем зеркально в список v: сосед — u
    p = (Edge*)malloc(sizeof(Edge));
    p->v = u; p->w = w; p->next = g->adj[v]; g->adj[v] = p;
    g->m++;
}
// Чтение графа из файла. Формат: первая строка — n; далее строки u v w
static int read_graph(const char* fname, Graph* g)
{
    FILE* f = fopen(fname, "r");
    if (!f) { printf("Oshibka\n"); return 0; }
    char line[256];
    int n = 0;
    int header_ok = 0;
    // Читаем первую непустую строку — число вершин
    while (fgets(line, sizeof(line), f)) {
        if (sscanf(line, "%d", &n) == 1) { header_ok = 1; break; }
    }
    if (!header_ok || n < 0) {
        printf("Oshibka\n");
        fclose(f); return 0;
    }
    g->n = n;
    g->m = 0;
    g->adj = (Edge**)calloc(n + 1, sizeof(Edge*));   // все NULL
    if (!g->adj) { printf("Oshibka\n"); fclose(f); return 0; }
    // Читаем рёбра: u v w
```

```
    while (fgets(line, sizeof(line), f)) {
        int u = 0, v = 0;
        long long w = 0;
        int got = sscanf(line, "%d %d %lld", &u, &v, &w);
        if (got < 3) continue;              // пустая строка или мусор
        // Вершина вне диапазона [1..n] — ошибка
        if (u < 1 || u > g->n || v < 1 || v > g->n) {
            printf("Oshibka\n");
            fclose(f); return 0;
        }
        if (w < 0) {
            printf("Oshibka\n");
            fclose(f); return 0;
        }
        // Петля — пропускаем (не влияет на кратчайшие пути)
        if (u == v) continue;
        add_edge(g, u, v, w);
    }
    fclose(f);
    return 1;
}
static void free_graph(Graph* g)
{
    for (int i = 1; i <= g->n; i++) {
        Edge* e = g->adj[i];
        while (e) { Edge* nx = e->next; free(e); e = nx; }
    }
    free(g->adj);
    g->adj = NULL;
    g->n = g->m = 0;
}
/*  АЛГОРИТМ ДЕЙКСТРЫ
 *  Ищет кратчайшие расстояния от вершины s до всех остальных.
 *  Работает только при неотрицательных весах.
 *  dist[i] — кратчайшее расстояние от s до i
 *  prev[i] — предшественник i на кратчайшем пути (для восстановления)
 *  visited[i] — 1, если вершина уже «зафиксирована» */
static void dijkstra(const Graph* g, int s, long long* dist, int* prev)
{
    int N = g->n;
    int* visited = (int*)calloc(N + 1, sizeof(int));
    // Инициализация
    for (int i = 1; i <= N; i++) { dist[i] = INF; prev[i] = 0; }
    dist[s] = 0;
    // N итераций — на каждой фиксируем одну вершину
    for (int step = 0; step < N; step++) {
        // 1) Находим непосещённую вершину с минимальным dist
        int u = -1;
        long long best = INF;
        for (int i = 1; i <= N; i++) {
            if (!visited[i] && dist[i] < best) { best = dist[i]; u = i;
}
        }
        if (u == -1) break;   // оставшиеся недостижимы
        // 2) Фиксируем её
        visited[u] = 1;
        // 3) Обновляем расстояния соседей
        for (const Edge* e = g->adj[u]; e; e = e->next) {
            int v = e->v;
```

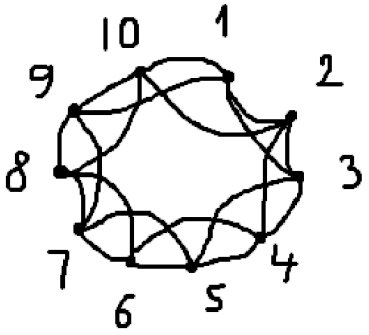
```
            if (!visited[v] && dist[u] + e->w < dist[v]) {
                dist[v] = dist[u] + e->w;
                prev[v] = u;
            }
        }
    }
    free(visited);
}
// Восстановление пути s → t по массиву prev
static void print_path(int* prev, int s, int t)
{
    if (t == s) { printf("%d", s); return; }
    if (prev[t] == 0) { printf("(net puti)"); return; }
    // Идём назад от t к s, складывая вершины в стек
    int stack[1024];
    int top = 0;
    int cur = t;
    while (cur != 0) {
        stack[top++] = cur;
        if (cur == s) break;
        cur = prev[cur];
    }
    // Выводим в обратном порядке
    for (int i = top - 1; i >= 0; i--) {
        printf("%d", stack[i]);
        if (i > 0) printf(" -> ");
    }
}
/*  АЛГОРИТМ ФЛОЙДА–УОРШЕЛЛА
 *  Ищет кратчайшие расстояния между ВСЕМИ парами вершин.
 *  Работает и с отрицательными весами (если нет отрицательных циклов).
 *  d[i][j] — кратчайшее расстояние от i до j */
static void floyd_warshall(const Graph* g, long long** d)
{
    int N = g->n;
    // Инициализация: 0 на диагонали, ∞ вне, веса прямых рёбер
    for (int i = 1; i <= N; i++)
        for (int j = 1; j <= N; j++)
            d[i][j] = (i == j) ? 0 : INF;
    // Прямые рёбра — берём минимальный вес (для кратных)
    for (int u = 1; u <= N; u++)
        for (const Edge* e = g->adj[u]; e; e = e->next)
            if (e->w < d[u][e->v]) d[u][e->v] = e->w;
    // Тройной цикл — разрешаем вершину k как промежуточную
    for (int k = 1; k <= N; k++)
        for (int i = 1; i <= N; i++)
            for (int j = 1; j <= N; j++)
                if (d[i][k] < INF && d[k][j] < INF) {
                    long long through = d[i][k] + d[k][j];
                    if (through < d[i][j]) d[i][j] = through;
                }
}
int main(int argc, char** argv)
{
    if (argc < 2) { printf("Oshibka\n"); return 1; }
    Graph g;
    if (!read_graph(argv[1], &g)) return 1;
    printf("Fayl: %s\n", argv[1]);
    printf("Vershin: %d, ryober: %d\n\n", g.n, g.m);
```

```
    int N = g.n;
    // ДЕЙКСТРА от вершины 1
    printf("=== ALGORITM DEYKSTRY (ot vershiny 1) ===\n");
    long long* dist = (long long*)malloc((N + 1) * sizeof(long long));
    int* prev = (int*)malloc((N + 1) * sizeof(int));
    if (!dist || !prev) {
        printf("Oshibka\n");
        free(dist); free(prev); free_graph(&g); return 1;
    }
    dijkstra(&g, 1, dist, prev);
    // Вывод расстояний
    printf("Krachaishie rasstoyaniya ot 1:\n");
    for (int i = 1; i <= N; i++) {
        printf("  do %2d: ", i);
        if (dist[i] >= INF) printf("nedostizhima\n");
        else printf("%lld\n", dist[i]);
    }
    // Вывод путей
    printf("\nKrachaishie puti ot 1:\n");
    for (int i = 1; i <= N; i++) {
        if (i == 1) continue;
        printf("  do %2d: ", i);
        if (dist[i] >= INF) printf("(net puti)\n");
        else { print_path(prev, 1, i); printf("   (dlina %lld)\n",
dist[i]); }
    }
    free(dist);
    free(prev);
    printf("\n");
    // ФЛОЙД-УОРШЕЛЛ
    printf("=== ALGORITM FLOYD-WARSHALL ===\n");
    long long** d = (long long**)malloc((N + 1) * sizeof(long long*));
    for (int i = 0; i <= N; i++)
        d[i] = (long long*)malloc((N + 1) * sizeof(long long));
    floyd_warshall(&g, d);
    // Вывод матрицы
    printf("Matritsa krachaishih rasstoyaniy:\n     ");
    for (int j = 1; j <= N; j++) printf("%6d", j);
    printf("\n");
    for (int i = 1; i <= N; i++) {
        printf("%4d:", i);
        for (int j = 1; j <= N; j++) {
            if (d[i][j] >= INF) printf("%6s", "INF");
            else printf("%6lld", d[i][j]);
        }
        printf("\n");
    }
    // Расстояния от 1 (для сверки с Дейкстрой)
    printf("\nKrachaishie rasstoyaniya ot 1 (Floyd):\n");
    for (int j = 1; j <= N; j++) {
        printf("  do %2d: ", j);
        if (d[1][j] >= INF) printf("nedostizhima\n");
        else printf("%lld\n", d[1][j]);
    }
    for (int i = 0; i <= N; i++) free(d[i]);
    free(d);
    free_graph(&g);
    return 0;
}
```

Проверки:

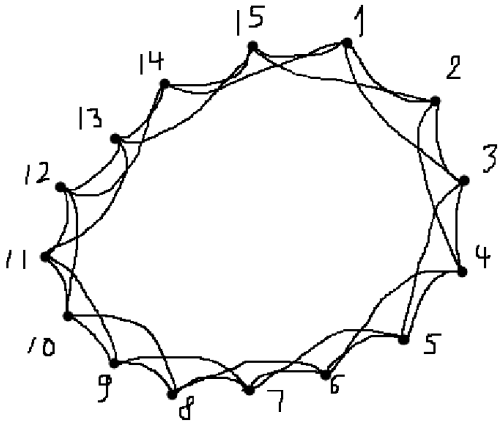
Тест01 - cirkulyant_10vershin

```
10
1 2 1536819634
1 3 -705715377972
2 3 -30
2 4 -673
3 4 85
3 5 0
4 5 -4274382698
4 6 -216
5 6 -2
5 7 5290970
6 7 1000000000000
6 8 -96334155009
7 8 -691376
7 9 38
8 9 -66159
8 10 -41452
9 1 -61535661690
9 10 34815457
10 1 -231
10 2 -5011120308
```



Oshibka

Тест02 - cirkulyant_15vershin




```
15
1 2 20
1 3 425165329989
2 3 37890
2 4 701
3 4 974754451
3 5 3453222462
4 5 10096
4 6 477641
5 6 735044390115
5 7 744
6 7 28676693
6 8 98
7 8 68536266820
7 9 6417963406
8 9 541
8 10 92
9 10 290341638342
9 11 95441357
10 11 173017
10 12 657876900
11 12 9904
11 13 4757753
12 13 98613689275
12 14 73596179260
13 14 65
13 15 1000000000000
14 1 227654057789
14 15 605966213
15 1 10632555939
15 2 672867
```

```
Fayl: tests/test02.txt
Vershin: 15, ryober: 30

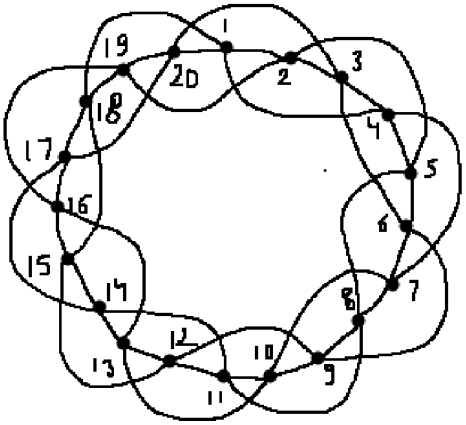
=== ALGORITM DEYKSTRY (ot vershiny 1) ===
Krachaishie rasstoyaniya ot 1:
  do  1: 0
  do  2: 20
  do  3: 37910
  do  4: 721
  do  5: 10817
  do  6: 478362
  do  7: 11561
  do  8: 478460
  do  9: 479001
  do 10: 478552
  do 11: 651569
  do 12: 661473
  do 13: 5409322
  do 14: 5409387
  do 15: 672887

Krachaishie puti ot 1:
  do  2: 1 -> 2   (dlina 20)
  do  3: 1 -> 2 -> 3   (dlina 37910)
  do  4: 1 -> 2 -> 4   (dlina 721)
  do  5: 1 -> 2 -> 4 -> 5   (dlina 10817)
  do  6: 1 -> 2 -> 4 -> 6   (dlina 478362)
  do  7: 1 -> 2 -> 4 -> 5 -> 7   (dlina 11561)
  do  8: 1 -> 2 -> 4 -> 6 -> 8   (dlina 478460)
  do  9: 1 -> 2 -> 4 -> 6 -> 8 -> 9   (dlina 479001)
  do 10: 1 -> 2 -> 4 -> 6 -> 8 -> 10   (dlina 478552)
  do 11: 1 -> 2 -> 4 -> 6 -> 8 -> 10 -> 11   (dlina 651569)
  do 12: 1 -> 2 -> 4 -> 6 -> 8 -> 10 -> 11 -> 12   (dlina 661473)
  do 13: 1 -> 2 -> 4 -> 6 -> 8 -> 10 -> 11 -> 13   (dlina 5409322)
  do 14: 1 -> 2 -> 4 -> 6 -> 8 -> 10 -> 11 -> 13 -> 14   (dlina 5409387)
  do 15: 1 -> 2 -> 15   (dlina 672887)
```

```
=== ALGORITM FLOYDIWARSHALL ===
Matritsa krachaishih rasstoyaniy:
          1           2           3           4           5           6           7           8           9          10          11          12          13          14          15
 1:         0          20       37910         721       10817      478362       11561      478460      479001      478552      651569      661473     5409322     5409387      672887
 2:        20           0       37890         701       10797      478342       11541      478440      478981      478532      651549      661453     5409302     5409367      672867
 3:     37910       37890           0       38591       48687      516232       49431      516330      516871      516422      689439      699343     5447192     5447257      710757
 4:       721         701       38591           0       10096      477641       10840      477739      478280      477831      650848      660752     5408601     5408666      673568
 5:     10817       10797       48687       10096           0      487737         744      487835      488376      487927      660944      670848     5418697     5418762      683664
 6:    478362      478342      516232      477641      487737           0      488481          98         639         190      173207      183111     4930960     4931025     1151209
 7:     11561       11541       49431       10840         744      488481           0      488579      489120      488671      661688      671592     5419441     5419506      684408
 8:    478460      478440      516330      477739      487835          98      488579           0         541          92      173189      183013     4930862     4930927     1151307
 9:    479001      478981      516871      478280      488376         639      489120         541           0         633      173650      183554     4931403     4931468     1151848
10:    478552      478532      516422      477831      487927         190      488671          92         633           0      173017      182921     4930770     4930835     1151399
11:    651569      651549      689439      650848      660944      173207      661688      173189      173650      173017           0        9904     4757753     4757818     1324416
12:    661473      661453      699343      660752      670848      183111      671592      183013      183554      182921        9904           0     4767657     4767722     1334320
13:   5409322     5409302     5447192     5408601     5418697     4930960     5419441     4930862     4931403     4930770     4757753     4767657           0          65     6082169
14:   5409387     5409367     5447257     5408666     5418762     4931025     5419506     4930927     4931468     4930835     4757818     4767722          65           0     6082234
15:    672887      672867      710757      673568      683664     1151209      684408     1151307     1151848     1151399     1324416     1334320     6082169     6082234           0

Krachaishie rasstoyaniya ot 1 (Floyd):
  do  1: 0
  do  2: 20
  do  3: 37910
  do  4: 721
  do  5: 10817
  do  6: 478362
  do  7: 11561
  do  8: 478460
  do  9: 479001
  do 10: 478552
  do 11: 651569
  do 12: 661473
  do 13: 5409322
  do 14: 5409387
  do 15: 672887
```

Тест03 - cirkulyant_20vershin



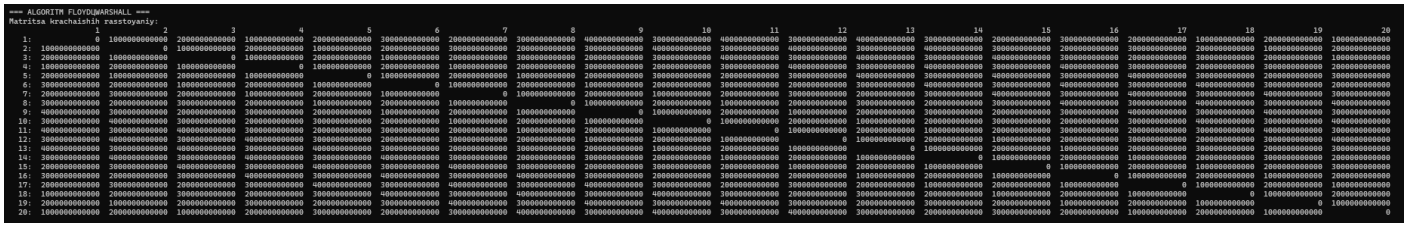

```
20
1 2 1000000000000
1 4 1000000000000
2 3 1000000000000
2 5 1000000000000
3 4 1000000000000
3 6 1000000000000
4 5 1000000000000
4 7 1000000000000
5 6 1000000000000
5 8 1000000000000
6 7 1000000000000
6 9 1000000000000
7 8 1000000000000
7 10 1000000000000
8 9 1000000000000
8 11 1000000000000
9 10 1000000000000
9 12 1000000000000
10 11 1000000000000
10 13 1000000000000
11 12 1000000000000
```

```
Fayl: tests/test03.txt
Vershin: 20, ryober: 40

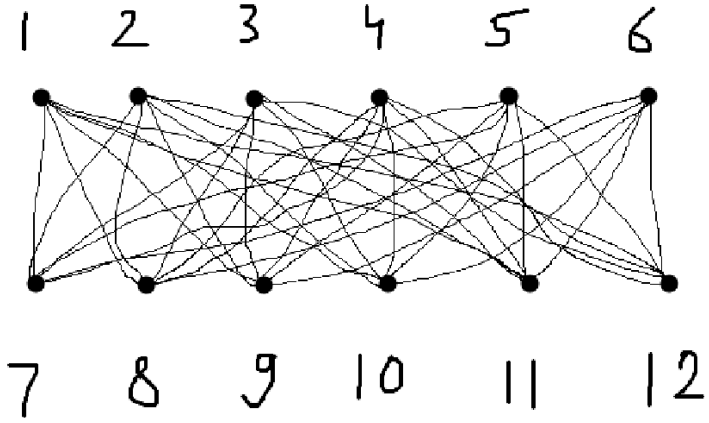
=== ALGORITM DEYKSTRY (ot vershiny 1) ===
Krachaishie rasstoyaniya ot 1:
  do  1: 0
  do  2: 1000000000000
  do  3: 2000000000000
  do  4: 1000000000000
  do  5: 2000000000000
  do  6: 3000000000000
  do  7: 2000000000000
  do  8: 3000000000000
  do  9: 4000000000000
  do 10: 3000000000000
  do 11: 4000000000000
  do 12: 3000000000000
  do 13: 4000000000000
  do 14: 3000000000000
  do 15: 2000000000000
  do 16: 3000000000000
  do 17: 2000000000000
  do 18: 1000000000000
  do 19: 2000000000000
  do 20: 1000000000000

Krachaishie puti ot 1:
  do  2: 1 -> 2   (dlina 1000000000000)
  do  3: 1 -> 2 -> 3   (dlina 2000000000000)
  do  4: 1 -> 4   (dlina 1000000000000)
  do  5: 1 -> 2 -> 5   (dlina 2000000000000)
  do  6: 1 -> 2 -> 3 -> 6   (dlina 3000000000000)
  do  7: 1 -> 4 -> 7   (dlina 2000000000000)
  do  8: 1 -> 2 -> 5 -> 8   (dlina 3000000000000)
  do  9: 1 -> 2 -> 3 -> 6 -> 9   (dlina 4000000000000)
  do 10: 1 -> 4 -> 7 -> 10   (dlina 3000000000000)
  do 11: 1 -> 2 -> 5 -> 8 -> 11   (dlina 4000000000000)
  do 12: 1 -> 18 -> 15 -> 12   (dlina 3000000000000)
  do 13: 1 -> 4 -> 7 -> 10 -> 13   (dlina 4000000000000)
  do 14: 1 -> 18 -> 15 -> 14   (dlina 3000000000000)
  do 15: 1 -> 18 -> 15   (dlina 2000000000000)
  do 16: 1 -> 18 -> 15 -> 16   (dlina 3000000000000)
  do 17: 1 -> 18 -> 17   (dlina 2000000000000)
  do 18: 1 -> 18   (dlina 1000000000000)
  do 19: 1 -> 2 -> 19   (dlina 2000000000000)
  do 20: 1 -> 20   (dlina 1000000000000)
```

```
Krachaishie rasstoyaniya ot 1 (Floyd):
  do  1: 0
  do  2: 1000000000000
  do  3: 2000000000000
  do  4: 1000000000000
  do  5: 2000000000000
  do  6: 3000000000000
  do  7: 2000000000000
  do  8: 3000000000000
  do  9: 4000000000000
  do 10: 3000000000000
  do 11: 4000000000000
  do 12: 3000000000000
  do 13: 4000000000000
  do 14: 3000000000000
  do 15: 2000000000000
  do 16: 3000000000000
  do 17: 2000000000000
  do 18: 1000000000000
  do 19: 2000000000000
  do 20: 1000000000000
```



Тест04 - K66_12vershin




```
12
1 7 1
1 8 2
1 9 3
1 10 4
1 11 5
1 12 6
2 7 7
2 8 8
2 9 9
2 10 10
2 11 11
2 12 12
3 7 13
3 8 14
3 9 15
```

```
Fayl: tests/test04.txt
Vershin: 12, ryober: 36

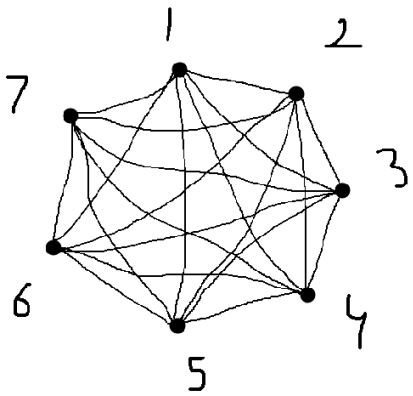
=== ALGORITM DEYKSTRY (ot vershiny 1) ===
Krachaishie rasstoyaniya ot 1:
  do  1: 0
  do  2: 8
  do  3: 14
  do  4: 20
  do  5: 26
  do  6: 32
  do  7: 1
  do  8: 2
  do  9: 3
  do 10: 4
  do 11: 5
  do 12: 6
```

```
Krachaishie puti ot 1:
  do  2: 1 -> 7 -> 2   (dlina 8)
  do  3: 1 -> 7 -> 3   (dlina 14)
  do  4: 1 -> 7 -> 4   (dlina 20)
  do  5: 1 -> 7 -> 5   (dlina 26)
  do  6: 1 -> 7 -> 6   (dlina 32)
  do  7: 1 -> 7   (dlina 1)
  do  8: 1 -> 8   (dlina 2)
  do  9: 1 -> 9   (dlina 3)
  do 10: 1 -> 10   (dlina 4)
  do 11: 1 -> 11   (dlina 5)
  do 12: 1 -> 12   (dlina 6)
```

```
=== ALGORITM FLOYD-WARSHALL ===
Matritsa krachaishih rasstoyaniy:
         1     2     3     4     5     6     7     8     9    10    11    12
  1:     0     8    14    20    26    32     1     2     3     4     5     6
  2:     8     0    20    26    32    38     7     8     9    10    11    12
  3:    14    20     0    32    38    44    13    14    15    16    17    18
  4:    20    26    32     0    44    50    19    20    21    22    23    24
  5:    26    32    38    44     0    56    25    26    27    28    29    30
  6:    32    38    44    50    56     0    31    32    33    34    35    36
  7:     1     7    13    19    25    31     0     3     4     5     6     7
  8:     2     8    14    20    26    32     3     0     5     6     7     8
  9:     3     9    15    21    27    33     4     5     0     7     8     9
 10:     4    10    16    22    28    34     5     6     7     0     9    10
 11:     5    11    17    23    29    35     6     7     8     9     0    11
 12:     6    12    18    24    30    36     7     8     9    10    11     0
```

```
Krachaishie rasstoyaniya ot 1 (Floyd):
  do  1: 0
  do  2: 8
  do  3: 14
  do  4: 20
  do  5: 26
  do  6: 32
  do  7: 1
  do  8: 2
  do  9: 3
  do 10: 4
  do 11: 5
  do 12: 6
```

Тест05 - K7_7vershin




```
7
1 2 1
1 3 2
1 4 3
1 5 4
1 6 5
1 7 6
2 3 1
2 4 1
2 5 1
```

```
Fayl: tests/test05.txt
Vershin: 7, ryober: 21

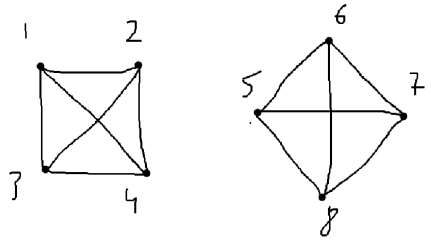
=== ALGORITM DEYKSTRY (ot vershiny 1) ===
Krachaishie rasstoyaniya ot 1:
  do  1: 0
  do  2: 1
  do  3: 2
  do  4: 2
  do  5: 2
  do  6: 2
  do  7: 2
```

```
Krachaishie puti ot 1:
  do  2: 1 -> 2   (dlina 1)
  do  3: 1 -> 3   (dlina 2)
  do  4: 1 -> 2 -> 4   (dlina 2)
  do  5: 1 -> 2 -> 5   (dlina 2)
  do  6: 1 -> 2 -> 6   (dlina 2)
  do  7: 1 -> 2 -> 7   (dlina 2)

=== ALGORITM FLOYD-WARSHALL ===
Matritsa krachaishih rasstoyaniy:
         1     2     3     4     5     6     7
  1:     0     1     2     2     2     2     2
  2:     1     0     1     1     1     1     1
  3:     2     1     0     1     1     1     1
  4:     2     1     1     0     1     1     1
  5:     2     1     1     1     0     1     1
  6:     2     1     1     1     1     0     1
  7:     2     1     1     1     1     1     0
```

```
Krachaishie rasstoyaniya ot 1 (Floyd):
  do  1: 0
  do  2: 1
  do  3: 2
  do  4: 2
  do  5: 2
  do  6: 2
  do  7: 2
```

Тест06 - eyler06_dva_K4_8vershin




```
8
1 2 3
1 3 4
1 4 5
2 3 5
2 4 6
3 4 7
5 6 3
5 7 4
5 8 5
6 7 5
6 8 6
7 8 7
```

```
Fayl: tests/test06.txt
Vershin: 8, ryober: 12

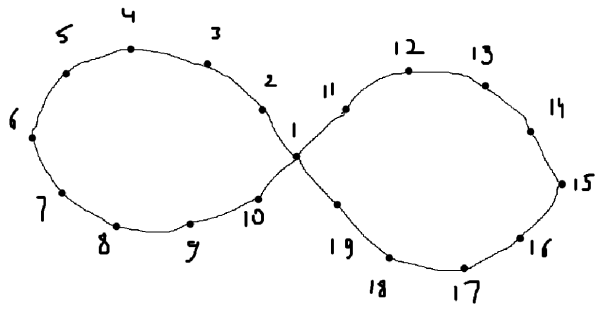
=== ALGORITM DEYKSTRY (ot vershiny 1) ===
Krachaishie rasstoyaniya ot 1:
  do  1: 0
  do  2: 3
  do  3: 4
  do  4: 5
  do  5: nedostizhima
  do  6: nedostizhima
  do  7: nedostizhima
  do  8: nedostizhima
```

```
Krachaishie puti ot 1:
  do  2: 1 -> 2   (dlina 3)
  do  3: 1 -> 3   (dlina 4)
  do  4: 1 -> 4   (dlina 5)
  do  5: (net puti)
  do  6: (net puti)
  do  7: (net puti)
  do  8: (net puti)
```

```
=== ALGORITM FLOYDWARSHALL ===
Matritsa krachaishih rasstoyaniy:
         1     2     3     4     5     6     7     8
  1:     0     3     4     5   INF   INF   INF   INF
  2:     3     0     5     6   INF   INF   INF   INF
  3:     4     5     0     7   INF   INF   INF   INF
  4:     5     6     7     0   INF   INF   INF   INF
  5:   INF   INF   INF   INF     0     3     4     5
  6:   INF   INF   INF   INF     3     0     5     6
  7:   INF   INF   INF   INF     4     5     0     7
  8:   INF   INF   INF   INF     5     6     7     0
```

```
Krachaishie rasstoyaniya ot 1 (Floyd):
  do  1: 0
  do  2: 3
  do  3: 4
  do  4: 5
  do  5: nedostizhima
  do  6: nedostizhima
  do  7: nedostizhima
  do  8: nedostizhima
```

Тест07 - pesochnye_chasy_19vershin




```
Fayl: tests/test07.txt
Vershin: 19, ryober: 20

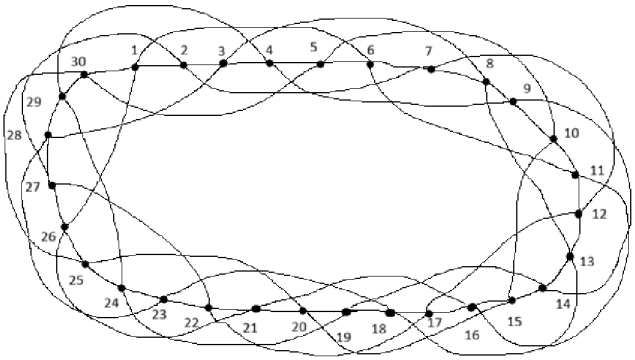
=== ALGORITM DEYKSTRY (ot vershiny 1) ===
Krachaishie rasstoyaniya ot 1:
  do  1: 0
  do  2: 1
  do  3: 2
  do  4: 3
  do  5: 4
  do  6: 5
  do  7: 6
  do  8: 7
  do  9: 8
  do 10: 9
  do 11: 9
  do 12: 8
  do 13: 7
  do 14: 6
  do 15: 5
  do 16: 4
  do 17: 3
  do 18: 2
  do 19: 1
```

```
Krachaishie puti ot 1:
  do  2: 1 -> 2   (dlina 1)
  do  3: 1 -> 2 -> 3   (dlina 2)
  do  4: 1 -> 2 -> 3 -> 4   (dlina 3)
  do  5: 1 -> 2 -> 3 -> 4 -> 5   (dlina 4)
  do  6: 1 -> 2 -> 3 -> 4 -> 5 -> 6   (dlina 5)
  do  7: 1 -> 2 -> 3 -> 4 -> 5 -> 6 -> 7   (dlina 6)
  do  8: 1 -> 2 -> 3 -> 4 -> 5 -> 6 -> 7 -> 8   (dlina 7)
  do  9: 1 -> 2 -> 3 -> 4 -> 5 -> 6 -> 7 -> 8 -> 9   (dlina 8)
  do 10: 1 -> 2 -> 3 -> 4 -> 5 -> 6 -> 7 -> 8 -> 9 -> 10   (dlina 9)
  do 11: 1 -> 19 -> 18 -> 17 -> 16 -> 15 -> 14 -> 13 -> 12 -> 11   (dlina 9)
  do 12: 1 -> 19 -> 18 -> 17 -> 16 -> 15 -> 14 -> 13 -> 12   (dlina 8)
  do 13: 1 -> 19 -> 18 -> 17 -> 16 -> 15 -> 14 -> 13   (dlina 7)
  do 14: 1 -> 19 -> 18 -> 17 -> 16 -> 15 -> 14   (dlina 6)
  do 15: 1 -> 19 -> 18 -> 17 -> 16 -> 15   (dlina 5)
  do 16: 1 -> 19 -> 18 -> 17 -> 16   (dlina 4)
  do 17: 1 -> 19 -> 18 -> 17   (dlina 3)
  do 18: 1 -> 19 -> 18   (dlina 2)
  do 19: 1 -> 19   (dlina 1)
```

```
=== ALGORITM FLOYD-WARSHALL ===
Matritsa krachaishih rasstoyaniy:
        1     2     3     4     5     6     7     8     9    10    11    12    13    14    15    16    17    18    19
  1:    0     1     2     3     4     5     6     7     8     9     9     8     7     6     5     4     3     2     1
  2:    1     0     1     2     3     4     5     6     7     8    10     9     8     7     6     5     4     3     2
  3:    2     1     0     1     2     3     4     5     6     7    11    10     9     8     7     6     5     4     3
  4:    3     2     1     0     1     2     3     4     5     6    12    11    10     9     8     7     6     5     4
  5:    4     3     2     1     0     1     2     3     4     5    13    12    11    10     9     8     7     6     5
  6:    5     4     3     2     1     0     1     2     3     4    14    13    12    11    10     9     8     7     6
  7:    6     5     4     3     2     1     0     1     2     3    15    14    13    12    11    10     9     8     7
  8:    7     6     5     4     3     2     1     0     1     2    16    15    14    13    12    11    10     9     8
  9:    8     7     6     5     4     3     2     1     0     1    17    16    15    14    13    12    11    10     9
 10:    9     8     7     6     5     4     3     2     1     0    18    17    16    15    14    13    12    11    10
 11:    9    10    11    12    13    14    15    16    17    18     0     1     2     3     4     5     6     7     8
 12:    8     9    10    11    12    13    14    15    16    17     1     0     1     2     3     4     5     6     7
 13:    7     8     9    10    11    12    13    14    15    16     2     1     0     1     2     3     4     5     6
 14:    6     7     8     9    10    11    12    13    14    15     3     2     1     0     1     2     3     4     5
 15:    5     6     7     8     9    10    11    12    13    14     4     3     2     1     0     1     2     3     4
 16:    4     5     6     7     8     9    10    11    12    13     5     4     3     2     1     0     1     2     3
 17:    3     4     5     6     7     8     9    10    11    12     6     5     4     3     2     1     0     1     2
 18:    2     3     4     5     6     7     8     9    10    11     7     6     5     4     3     2     1     0     1
 19:    1     2     3     4     5     6     7     8     9    10     8     7     6     5     4     3     2     1     0
```

```
Krachaishie rasstoyaniya ot 1 (Floyd):
  do  1: 0
  do  2: 1
  do  3: 2
  do  4: 3
  do  5: 4
  do  6: 5
  do  7: 6
  do  8: 7
  do  9: 8
  do 10: 9
  do 11: 9
  do 12: 8
  do 13: 7
  do 14: 6
  do 15: 5
  do 16: 4
  do 17: 3
  do 18: 2
  do 19: 1
```

Тест08 – cirkulyant _30vershin




```
Fayl: tests/test08.txt
Vershin: 30, ryober: 60

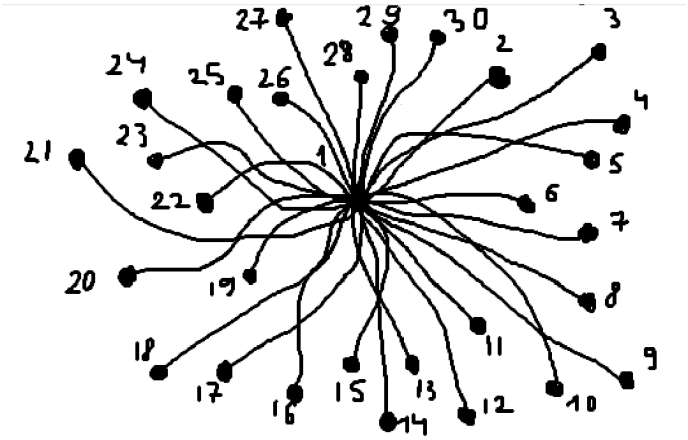
=== ALGORITM DEYKSTRY (ot vershiny 1) ===
Krachaishie rasstoyaniya ot 1:
  do  1: 0
  do  2: 1000
  do  3: 2000
  do  4: 3000
```

```
Krachaishie puti ot 1:
  do  2: 1 -> 2   (dlina 1000)
  do  3: 1 -> 2 -> 3   (dlina 2000)
  do  4: 1 -> 2 -> 3 -> 4   (dlina 3000)
  do  5: 1 -> 6 -> 5   (dlina 2000)
  do  6: 1 -> 6   (dlina 1000)
  do  7: 1 -> 2 -> 7   (dlina 2000)
  do  8: 1 -> 2 -> 3 -> 8   (dlina 3000)
  do  9: 1 -> 2 -> 3 -> 4 -> 9   (dlina 4000)
  do 10: 1 -> 6 -> 5 -> 10   (dlina 3000)
  do 11: 1 -> 6 -> 11   (dlina 2000)
  do 12: 1 -> 2 -> 7 -> 12   (dlina 3000)
  do 13: 1 -> 2 -> 3 -> 8 -> 13   (dlina 4000)
  do 14: 1 -> 2 -> 3 -> 4 -> 9 -> 14   (dlina 5000)
  do 15: 1 -> 6 -> 5 -> 10 -> 15   (dlina 4000)
  do 16: 1 -> 6 -> 11 -> 16   (dlina 3000)
  do 17: 1 -> 2 -> 7 -> 12 -> 17   (dlina 4000)
  do 18: 1 -> 2 -> 3 -> 8 -> 13 -> 18   (dlina 5000)
  do 19: 1 -> 26 -> 21 -> 20 -> 19   (dlina 4000)
  do 20: 1 -> 26 -> 21 -> 20   (dlina 3000)
  do 21: 1 -> 26 -> 21   (dlina 2000)
  do 22: 1 -> 26 -> 21 -> 22   (dlina 3000)
  do 23: 1 -> 26 -> 21 -> 22 -> 23   (dlina 4000)
  do 24: 1 -> 26 -> 25 -> 24   (dlina 3000)
  do 25: 1 -> 26 -> 25   (dlina 2000)
  do 26: 1 -> 26   (dlina 1000)
  do 27: 1 -> 2 -> 27   (dlina 2000)
  do 28: 1 -> 2 -> 3 -> 28   (dlina 3000)
  do 29: 1 -> 30 -> 29   (dlina 2000)
  do 30: 1 -> 30   (dlina 1000)
```

```
Krachaishie rasstoyaniya ot 1 (Floyd):
  do  1: 0
  do  2: 1000
  do  3: 2000
  do  4: 3000
  do  5: 2000
  do  6: 1000
  do  7: 2000
  do  8: 3000
  do  9: 4000
  do 10: 3000
  do 11: 2000
  do 12: 3000
  do 13: 4000
  do 14: 5000
  do 15: 4000
  do 16: 3000
  do 17: 4000
  do 18: 5000
  do 19: 4000
  do 20: 3000
  do 21: 2000
  do 22: 3000
  do 23: 4000
  do 24: 3000
  do 25: 2000
  do 26: 1000
  do 27: 2000
  do 28: 3000
  do 29: 2000
  do 30: 1000
```

```
=== ALGORITM FLOYDUWARSHALL ===
Matritsa krachaishih rasstoyaniy:
        1    2    3    4    5    6    7    8    9   10   11   12   13   14   15   16   17   18   19   20   21   22   23   24   25   26   27   28   29   30
   1:    0 1000 2000 3000 2000 1000 2000 3000 4000 3000 2000 3000 4000 5000 4000 3000 4000 5000 4000 3000 2000 3000 4000 3000 2000 1000 2000 3000 2000 1000
   2: 1000    0 1000 2000 3000 2000 1000 2000 3000 4000 3000 2000 3000 4000 5000 4000 3000 4000 5000 4000 3000 2000 3000 4000 3000 2000 1000 2000 3000 2000
   3: 2000 1000    0 1000 2000 3000 2000 1000 2000 3000 4000 3000 2000 3000 4000 5000 4000 3000 4000 5000 4000 3000 2000 3000 4000 3000 2000 1000 2000 3000
   4: 3000 2000 1000    0 1000 2000 3000 2000 1000 2000 3000 4000 3000 2000 3000 4000 5000 4000 3000 4000 5000 4000 3000 2000 3000 4000 3000 2000 1000 2000
   5: 2000 3000 2000 1000    0 1000 2000 3000 2000 1000 2000 3000 4000 3000 2000 3000 4000 5000 4000 3000 4000 5000 4000 3000 2000 3000 4000 3000 2000 1000
   6: 1000 2000 3000 2000 1000    0 1000 2000 3000 2000 1000 2000 3000 4000 3000 2000 3000 4000 5000 4000 3000 4000 5000 4000 3000 2000 3000 4000 3000 2000
   7: 2000 1000 2000 3000 2000 1000    0 1000 2000 3000 2000 1000 2000 3000 4000 3000 2000 3000 4000 5000 4000 3000 4000 5000 4000 3000 2000 3000 4000 3000
   8: 3000 2000 1000 2000 3000 2000 1000    0 1000 2000 3000 2000 1000 2000 3000 4000 3000 2000 3000 4000 5000 4000 3000 4000 5000 4000 3000 2000 3000 4000
   9: 4000 3000 2000 1000 2000 3000 2000 1000    0 1000 2000 3000 2000 1000 2000 3000 4000 3000 2000 3000 4000 5000 4000 3000 4000 5000 4000 3000 2000 3000
  10: 3000 4000 3000 2000 1000 2000 3000 2000 1000    0 1000 2000 3000 2000 1000 2000 3000 4000 3000 2000 3000 4000 5000 4000 3000 4000 5000 4000 3000 2000
  11: 2000 3000 4000 3000 2000 1000 2000 3000 2000 1000    0 1000 2000 3000 2000 1000 2000 3000 4000 3000 2000 3000 4000 5000 4000 3000 4000 5000 4000 3000
  12: 3000 2000 3000 4000 3000 2000 1000 2000 3000 2000 1000    0 1000 2000 3000 2000 1000 2000 3000 4000 3000 2000 3000 4000 5000 4000 3000 4000 5000 4000
  13: 4000 3000 2000 3000 4000 3000 2000 1000 2000 3000 2000 1000    0 1000 2000 3000 2000 1000 2000 3000 4000 3000 2000 3000 4000 5000 4000 3000 4000 5000
  14: 5000 4000 3000 2000 3000 4000 3000 2000 1000 2000 3000 2000 1000    0 1000 2000 3000 2000 1000 2000 3000 4000 3000 2000 3000 4000 5000 4000 3000 4000
  15: 4000 5000 4000 3000 2000 3000 4000 3000 2000 1000 2000 3000 2000 1000    0 1000 2000 3000 2000 1000 2000 3000 4000 3000 2000 3000 4000 5000 4000 3000
  16: 3000 4000 5000 4000 3000 2000 3000 4000 3000 2000 1000 2000 3000 2000 1000    0 1000 2000 3000 2000 1000 2000 3000 4000 3000 2000 3000 4000 5000 4000
  17: 4000 3000 4000 5000 4000 3000 2000 3000 4000 3000 2000 1000 2000 3000 2000 1000    0 1000 2000 3000 2000 1000 2000 3000 4000 3000 2000 3000 4000 5000
  18: 5000 4000 3000 4000 5000 4000 3000 2000 3000 4000 3000 2000 1000 2000 3000 2000 1000    0 1000 2000 3000 2000 1000 2000 3000 4000 3000 2000 3000 4000
  19: 4000 5000 4000 3000 4000 5000 4000 3000 2000 3000 4000 3000 2000 1000 2000 3000 2000 1000    0 1000 2000 3000 2000 1000 2000 3000 4000 3000 2000 3000
  20: 3000 4000 5000 4000 3000 4000 5000 4000 3000 2000 3000 4000 3000 2000 1000 2000 3000 2000 1000    0 1000 2000 3000 2000 1000 2000 3000 4000 3000 2000
  21: 2000 3000 4000 5000 4000 3000 4000 5000 4000 3000 2000 3000 4000 3000 2000 1000 2000 3000 2000 1000    0 1000 2000 3000 2000 1000 2000 3000 4000 3000
  22: 3000 2000 3000 4000 5000 4000 3000 4000 5000 4000 3000 2000 3000 4000 3000 2000 1000 2000 3000 2000 1000    0 1000 2000 3000 2000 1000 2000 3000 4000
  23: 4000 3000 2000 3000 4000 5000 4000 3000 4000 5000 4000 3000 2000 3000 4000 3000 2000 1000 2000 3000 2000 1000    0 1000 2000 3000 2000 1000 2000 3000
  24: 3000 4000 3000 2000 3000 4000 5000 4000 3000 4000 5000 4000 3000 2000 3000 4000 3000 2000 1000 2000 3000 2000 1000    0 1000 2000 3000 2000 1000 2000
  25: 2000 3000 4000 3000 2000 3000 4000 5000 4000 3000 4000 5000 4000 3000 2000 3000 4000 3000 2000 1000 2000 3000 2000 1000    0 1000 2000 3000 2000 1000
  26: 1000 2000 3000 4000 3000 2000 3000 4000 5000 4000 3000 4000 5000 4000 3000 2000 3000 4000 3000 2000 1000 2000 3000 2000 1000    0 1000 2000 3000 2000
  27: 2000 1000 2000 3000 4000 3000 2000 3000 4000 5000 4000 3000 4000 5000 4000 3000 2000 3000 4000 3000 2000 1000 2000 3000 2000 1000    0 1000 2000 3000
  28: 3000 2000 1000 2000 3000 4000 3000 2000 3000 4000 5000 4000 3000 4000 5000 4000 3000 2000 3000 4000 3000 2000 1000 2000 3000 2000 1000    0 1000 2000
  29: 2000 3000 2000 1000 2000 3000 4000 3000 2000 3000 4000 5000 4000 3000 4000 5000 4000 3000 2000 3000 4000 3000 2000 1000 2000 3000 2000 1000    0 1000
  30: 1000 2000 3000 2000 1000 2000 3000 4000 3000 2000 3000 4000 5000 4000 3000 4000 5000 4000 3000 2000 3000 4000 3000 2000 1000 2000 3000 2000 1000    0
```

Тест09 – zvezda_30vershin



```
Fayl: tests/test09.txt
Vershin: 30, ryober: 29

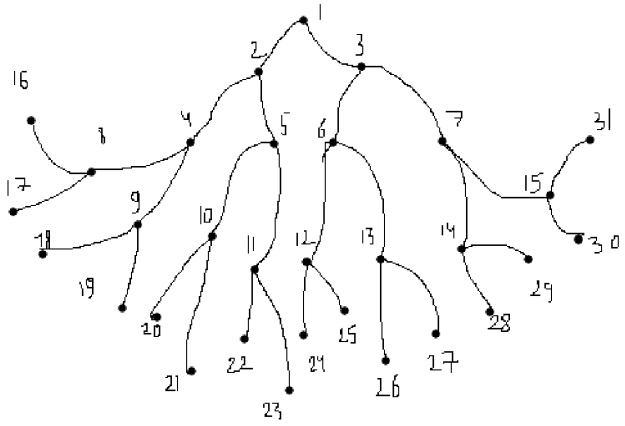
=== ALGORITM DEYKSTRY (ot vershiny 1) ===
Krachaishie rasstoyaniya ot 1:
  do  1: 0
  do  2: 59453716
  do  3: 45
  do  4: 823
  do  5: 38900975
  do  6: 1152
  do  7: 904
  do  8: 736867156
  do  9: 95530
  do 10: 2
  do 11: 5
  do 12: 54
  do 13: 6655
  do 14: 9969337655
  do 15: 419515662813
  do 16: 40218146230
  do 17: 713
  do 18: 852978062
  do 19: 740318
  do 20: 1490370
  do 21: 39191322174
  do 22: 9
  do 23: 415558381
  do 24: 9051739
  do 25: 65811281236
  do 26: 56304762
  do 27: 4183409495
  do 28: 30932
  do 29: 350151790144
  do 30: 7
```

```
Krachaishie puti ot 1:
  do  2: 1 -> 2   (dlina 59453716)
  do  3: 1 -> 3   (dlina 45)
  do  4: 1 -> 4   (dlina 823)
  do  5: 1 -> 5   (dlina 38900975)
  do  6: 1 -> 6   (dlina 1152)
  do  7: 1 -> 7   (dlina 904)
  do  8: 1 -> 8   (dlina 736867156)
  do  9: 1 -> 9   (dlina 95530)
  do 10: 1 -> 10   (dlina 2)
  do 11: 1 -> 11   (dlina 5)
  do 12: 1 -> 12   (dlina 54)
  do 13: 1 -> 13   (dlina 6655)
  do 14: 1 -> 14   (dlina 9969337655)
  do 15: 1 -> 15   (dlina 419515662813)
  do 16: 1 -> 16   (dlina 40218146230)
  do 17: 1 -> 17   (dlina 713)
  do 18: 1 -> 18   (dlina 852978062)
  do 19: 1 -> 19   (dlina 740318)
  do 20: 1 -> 20   (dlina 1490370)
  do 21: 1 -> 21   (dlina 39191322174)
  do 22: 1 -> 22   (dlina 9)
  do 23: 1 -> 23   (dlina 415558381)
  do 24: 1 -> 24   (dlina 9051739)
  do 25: 1 -> 25   (dlina 65811281236)
  do 26: 1 -> 26   (dlina 56304762)
  do 27: 1 -> 27   (dlina 4183409495)
  do 28: 1 -> 28   (dlina 30932)
  do 29: 1 -> 29   (dlina 350151790144)
  do 30: 1 -> 30   (dlina 7)
```



```
Krachaishie rasstoyaniya ot 1 (Floyd):
  do  1: 0
  do  2: 59453716
  do  3: 45
  do  4: 823
  do  5: 38900975
  do  6: 1152
  do  7: 904
  do  8: 736867156
  do  9: 95530
  do 10: 2
  do 11: 5
  do 12: 54
  do 13: 6655
  do 14: 9969337655
  do 15: 419515662813
  do 16: 40218146230
  do 17: 713
  do 18: 852978062
  do 19: 740318
  do 20: 1490370
  do 21: 39191322174
  do 22: 9
  do 23: 415558381
  do 24: 9051739
  do 25: 65811281236
  do 26: 56304762
  do 27: 4183409495
  do 28: 30932
  do 29: 350151790144
  do 30: 7
```

Тест10 – derevo_31vershin



```
Fayl: tests/test10.txt
Vershin: 31, ryober: 30

=== ALGORITM DEYKSTRY (ot vershiny 1) ===
Krachaishie rasstoyaniya ot 1:
  do  1: 0
  do  2: 985045255
  do  3: 605144438
  do  4: 989085381
  do  5: 985045255
  do  6: 672399354
  do  7: 84637554173
  do  8: 989085764
  do  9: 989094890
  do 10: 985045705
  do 11: 985053005
  do 12: 672404271
  do 13: 7789861786
  do 14: 84744091905
  do 15: 84963308168
  do 16: 989146931
  do 17: 1077754917
  do 18: 1063884359
  do 19: 298599953676
  do 20: 992891613
  do 21: 985418654
  do 22: 1006896275
  do 23: 93994931802
  do 24: 672495720
  do 25: 672404467
  do 26: 7790119578
  do 27: 7789862466
  do 28: 84747675034
  do 29: 84744091999
  do 30: 84968914866
  do 31: 84963308631
```

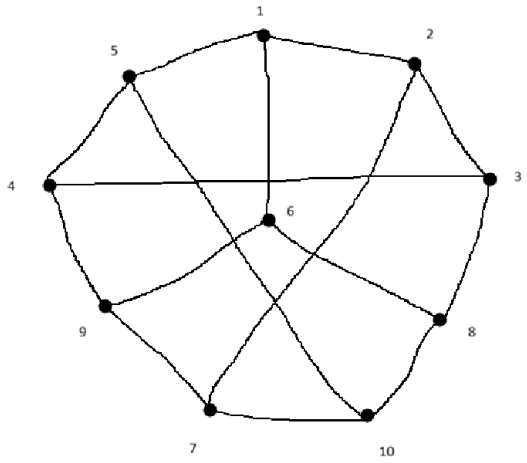
```
Krachaishie puti ot 1:
  do  2: 1 -> 2   (dlina 985045255)
  do  3: 1 -> 3   (dlina 605144438)
  do  4: 1 -> 2 -> 4   (dlina 989085381)
  do  5: 1 -> 2 -> 5   (dlina 985045255)
  do  6: 1 -> 3 -> 6   (dlina 672399354)
  do  7: 1 -> 3 -> 7   (dlina 84637554173)
  do  8: 1 -> 2 -> 4 -> 8   (dlina 989085764)
  do  9: 1 -> 2 -> 4 -> 9   (dlina 989094890)
  do 10: 1 -> 2 -> 5 -> 10   (dlina 985045705)
  do 11: 1 -> 2 -> 5 -> 11   (dlina 985053005)
  do 12: 1 -> 3 -> 6 -> 12   (dlina 672404271)
  do 13: 1 -> 3 -> 6 -> 13   (dlina 7789861786)
  do 14: 1 -> 3 -> 7 -> 14   (dlina 84744091905)
  do 15: 1 -> 3 -> 7 -> 15   (dlina 84963308168)
  do 16: 1 -> 2 -> 4 -> 8 -> 16   (dlina 989146931)
  do 17: 1 -> 2 -> 4 -> 8 -> 17   (dlina 1077754917)
  do 18: 1 -> 2 -> 4 -> 9 -> 18   (dlina 1063884359)
  do 19: 1 -> 2 -> 4 -> 9 -> 19   (dlina 298599953676)
  do 20: 1 -> 2 -> 5 -> 10 -> 20   (dlina 992891613)
  do 21: 1 -> 2 -> 5 -> 10 -> 21   (dlina 985418654)
  do 22: 1 -> 2 -> 5 -> 11 -> 22   (dlina 1006896275)
  do 23: 1 -> 2 -> 5 -> 11 -> 23   (dlina 93994931802)
  do 24: 1 -> 3 -> 6 -> 12 -> 24   (dlina 672495720)
  do 25: 1 -> 3 -> 6 -> 12 -> 25   (dlina 672404467)
  do 26: 1 -> 3 -> 6 -> 13 -> 26   (dlina 7790119578)
  do 27: 1 -> 3 -> 6 -> 13 -> 27   (dlina 7789862466)
  do 28: 1 -> 3 -> 7 -> 14 -> 28   (dlina 84747675034)
  do 29: 1 -> 3 -> 7 -> 14 -> 29   (dlina 84744091999)
  do 30: 1 -> 3 -> 7 -> 15 -> 30   (dlina 84968914866)
  do 31: 1 -> 3 -> 7 -> 15 -> 31   (dlina 84963308631)
```



```
Krachaishie rasstoyaniya ot 1 (Floyd):
  do  1: 0
  do  2: 985045255
  do  3: 605144438
  do  4: 989085381
  do  5: 985045255
  do  6: 672399354
  do  7: 84637554173
  do  8: 989085764
  do  9: 989094890
  do 10: 985045705
  do 11: 985053005
  do 12: 672404271
  do 13: 7789861786
  do 14: 84744091905
  do 15: 84963308168
  do 16: 989146931
  do 17: 1077754917
```

```
do 18: 1063884359
do 19: 298599953676
do 20: 992891613
do 21: 985418654
do 22: 1006896275
do 23: 93994931802
do 24: 672495720
do 25: 672404467
do 26: 7790119578
do 27: 7789862466
do 28: 84747675034
do 29: 84744091999
do 30: 84968914866
do 31: 84963308631
```

Тест11 – Petersen_10vershin




```
10
1 2 43170240988
2 3 74
3 4 48
4 5 424159022
5 1 6292047
6 8 2475
8 10 53173080415
10 7 27767652
7 9 4591549
9 6 840674872437
1 6 79
2 7 574
3 8 78917692
4 9 91924781384
5 10 363709366
```

```
Fayl: tests/test11.txt
Vershin: 10, ryober: 15

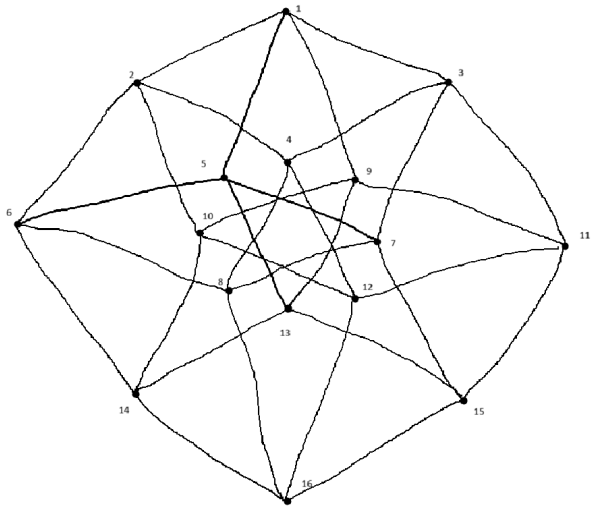
=== ALGORITM DEYKSTRY (ot vershiny 1) ===
Krachaishie rasstoyaniya ot 1:
  do  1: 0
  do  2: 78920320
  do  3: 78920246
  do  4: 78920294
  do  5: 6292047
  do  6: 79
  do  7: 78920894
  do  8: 2554
  do  9: 83512443
  do 10: 106688546

Krachaishie puti ot 1:
  do  2: 1 -> 6 -> 8 -> 3 -> 2   (dlina 78920320)
  do  3: 1 -> 6 -> 8 -> 3   (dlina 78920246)
  do  4: 1 -> 6 -> 8 -> 3 -> 4   (dlina 78920294)
  do  5: 1 -> 5   (dlina 6292047)
  do  6: 1 -> 6   (dlina 79)
  do  7: 1 -> 6 -> 8 -> 3 -> 2 -> 7   (dlina 78920894)
  do  8: 1 -> 6 -> 8   (dlina 2554)
  do  9: 1 -> 6 -> 8 -> 3 -> 2 -> 7 -> 9   (dlina 83512443)
  do 10: 1 -> 6 -> 8 -> 3 -> 2 -> 7 -> 10   (dlina 106688546)
```

```
Krachaishie rasstoyaniya ot 1 (Floyd):
  do  1: 0
  do  2: 78920320
  do  3: 78920246
  do  4: 78920294
  do  5: 6292047
  do  6: 79
  do  7: 78920894
  do  8: 2554
  do  9: 83512443
  do 10: 106688546
```

```
=== ALGORITM FLOYDUWARSHALL ===
Matritsa krachaishih rasstoyaniy:
          1          2          3          4          5          6          7          8          9         10
 1:       0   78920320   78920246   78920294    6292047         79   78920894       2554   83512443  106688546
 2:78920320          0         74        122   85212367   78920241        574   78917766    4592123   27768226
 3:78920246         74          0         48   85212293   78920167        648   78917692    4592197   27768300
 4:78920294        122         48          0   85212341   78920215        696   78917740    4592245   27768348
 5: 6292047   85212367   85212293   85212341          0    6292126   85212941    6294601   89804490  112980593
 6:      79   78920241   78920167   78920215    6292126          0   78920815       2475   83512364  106688467
 7:78920894        574        648        696   85212941   78920815          0   78918340    4591549   27767652
 8:    2554   78917766   78917692   78917740    6294601       2475   78918340          0   83509889  106685992
 9:83512443    4592123    4592197    4592245   89804490   83512364    4591549   83509889          0   32359201
10:106688546   27768226   27768300   27768348  112980593  106688467   27767652  106685992   32359201          0
```

Тест12 – giperkub_Q4_16vershin




```
16
1 2 2
1 3 3
1 5 5
1 9 9
2 4 8
2 6 12
2 10 20
3 4 12
3 7 21
3 11 33
4 8 32
4 12 48
5 6 30
5 7 35
5 13 65
6 8 48
6 14 84
7 8 56
7 15 105
8 16 128
9 10 90
9 11 99
9 13 117
10 12 120
10 14 140
11 12 132
11 15 165
12 16 184
13 14 182
13 15 195
14 16 224
15 16 240
```

```
Fayl: tests/test12.txt
Vershin: 16, ryober: 32

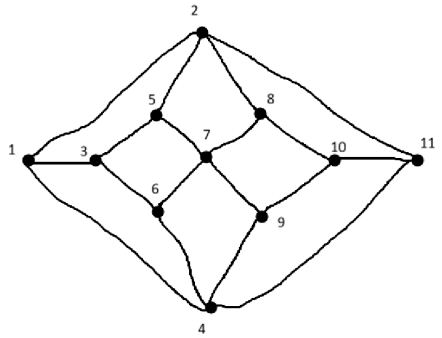
=== ALGORITM DEYKSTRY (ot vershiny 1) ===
Krachaishie rasstoyaniya ot 1:
  do  1: 0
  do  2: 2
  do  3: 3
  do  4: 10
  do  5: 5
  do  6: 14
  do  7: 24
  do  8: 42
  do  9: 9
  do 10: 22
  do 11: 36
  do 12: 58
  do 13: 70
  do 14: 98
  do 15: 129
  do 16: 170
```

```
Krachaishie puti ot 1:
  do  2: 1 -> 2   (dlina 2)
  do  3: 1 -> 3   (dlina 3)
  do  4: 1 -> 2 -> 4   (dlina 10)
  do  5: 1 -> 5   (dlina 5)
  do  6: 1 -> 2 -> 6   (dlina 14)
  do  7: 1 -> 3 -> 7   (dlina 24)
  do  8: 1 -> 2 -> 4 -> 8   (dlina 42)
  do  9: 1 -> 9   (dlina 9)
  do 10: 1 -> 2 -> 10   (dlina 22)
  do 11: 1 -> 3 -> 11   (dlina 36)
  do 12: 1 -> 2 -> 4 -> 12   (dlina 58)
  do 13: 1 -> 5 -> 13   (dlina 70)
  do 14: 1 -> 2 -> 6 -> 14   (dlina 98)
  do 15: 1 -> 3 -> 7 -> 15   (dlina 129)
  do 16: 1 -> 2 -> 4 -> 8 -> 16   (dlina 170)
```

```
Krachaishie rasstoyaniya ot 1 (Floyd):
  do  1: 0
  do  2: 2
  do  3: 3
  do  4: 10
  do  5: 5
  do  6: 14
  do  7: 24
  do  8: 42
  do  9: 9
  do 10: 22
  do 11: 36
  do 12: 58
  do 13: 70
  do 14: 98
  do 15: 129
  do 16: 170
```

```
=== ALGORITM FLOYDWARSHALL ===
Matritsa krachaishih rasstoyaniy:
          1         2         3         4         5         6         7         8         9        10        11        12        13        14        15        16
 1:       0         2         3        10         5        14        24        42         9        22        36        58        70        98       129       170
 2:       2         0         5         8         7        12        26        40        11        20        38        56        72        96       131       168
 3:       3         5         0        12         8        17        21        44        12        25        33        60        73       101       126       172
 4:      10         8        12         0        15        20        33        32        19        28        45        48        80       104       138       160
 5:       5         7         8        15         0        19        29        47        14        27        41        63        65       103       134       175
 6:      14        12        17        20        19         0        38        48        23        32        50        68        84        84       143       176
 7:      24        26        21        33        29        38         0        56        33        46        54        81        94       122       105       184
 8:      42        40        44        32        47        48        56         0        51        60        77        80       112       132       161       128
 9:       9        11        12        19        14        23        33        51         0        31        45        67        79       107       138       179
10:      22        20        25        28        27        32        46        60        31         0        58        76        92       116       151       188
11:      36        38        33        45        41        50        54        77        45        58         0        93       106       134       159       205
12:      58        56        60        48        63        68        81        80        67        76        93         0       128       152       186       184
13:      70        72        73        80        65        84        94       112        79        92       106       128         0       168       195       240
14:      98        96       101       104       103        84       122       132       107       116       134       152       168         0       227       224
15:     129       131       126       138       134       143       105       161       138       151       159       186       195       227         0       240
16:     170       168       172       160       175       176       184       128       179       188       205       184       240       224       240         0
```

Тест13 – Herschel_11vershin




```
Fayl: tests/test13.txt
Vershin: 11, ryober: 18

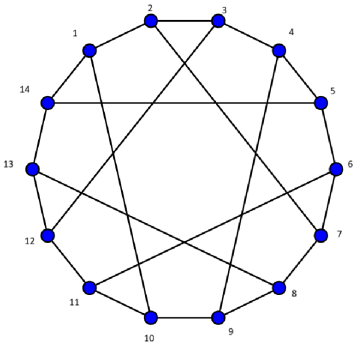
=== ALGORITM DEYKSTRY (ot vershiny 1) ===
Krachaishie rasstoyaniya ot 1:
  do  1: 0
  do  2: 7856523
  do  3: 2242106990
  do  4: 13990064
  do  5: 8570574
  do  6: 82774202
  do  7: 7950881
  do  8: 7856612
  do  9: 7950143
  do 10: 7949683
  do 11: 13989987

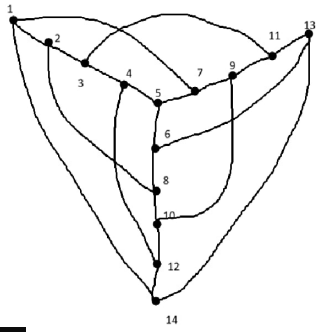
Krachaishie puti ot 1:
  do  2: 1 -> 2   (dlina 7856523)
  do  3: 1 -> 3   (dlina 2242106990)
  do  4: 1 -> 2 -> 11 -> 4   (dlina 13990064)
  do  5: 1 -> 2 -> 5   (dlina 8570574)
  do  6: 1 -> 2 -> 8 -> 10 -> 9 -> 7 -> 6   (dlina 82774202)
  do  7: 1 -> 2 -> 8 -> 10 -> 9 -> 7   (dlina 7950881)
  do  8: 1 -> 2 -> 8   (dlina 7856612)
  do  9: 1 -> 2 -> 8 -> 10 -> 9   (dlina 7950143)
  do 10: 1 -> 2 -> 8 -> 10   (dlina 7949683)
  do 11: 1 -> 2 -> 11   (dlina 13989987)
```

```
Krachaishie rasstoyaniya ot 1 (Floyd):
  do  1: 0
  do  2: 7856523
  do  3: 2242106990
  do  4: 13990064
  do  5: 8570574
  do  6: 82774202
  do  7: 7950881
  do  8: 7856612
  do  9: 7950143
  do 10: 7949683
  do 11: 13989987
```

```
=== ALGORITM FLOYD–WARSHALL ===
Matritsa krachaishih rasstoyaniy:
          1          2          3          4          5          6          7          8          9         10         11
 1:       0    7856523 2242106990   13990064    8570574   82774202    7950881    7856612    7950143    7949683   13989987
 2:  7856523          0 2249963513    6133541     714051   74917679      94358         89      93620      93160    6133464
 3: 2242106990 2249963513          0 2256097054 2250677564 2324881192 2250057871 2249963602 2250057133 2250056673 2256096977
 4:   13990064    6133541 2256097054          0    6847592   81051220    6227899    6133630    6227161    6226701         77
 5:    8570574     714051 2250677564    6847592          0   75631730     808409     714140     807671     807211    6847515
 6:   82774202   74917679 2324881192   81051220   75631730          0   74823321   74917590   74824059   74824519   81051143
 7:    7950881      94358 2250057871    6227899     808409   74823321          0      94269        738       1198    6227822
 8:    7856612         89 2249963602    6133630     714140   74917590      94269          0      93531      93071    6133553
 9:    7950143      93620 2250057133    6227161     807671   74824059        738      93531          0        460    6227084
10:    7949683      93160 2250056673    6226701     807211   74824519       1198      93071        460          0    6226624
11:   13989987    6133464 2256096977         77    6847515   81051143    6227822    6133553    6227084    6226624          0
```

Тест14 – Hivud_14vershin





```
14
1 2 21
1 10 11
1 14 411
2 3 32
2 7 72
3 4 43
3 12 213
4 5 54
4 9 94
5 6 65
5 14 415
6 7 76
6 11 116
7 8 87
8 9 98
8 13 318
9 10 19
10 11 1101
11 12 2111
12 13 3121
13 14 4131
```

```
Fayl: tests/test14.txt
Vershin: 14, ryober: 21

=== ALGORITM DEYKSTRY (ot vershiny 1) ===
Krachaishie rasstoyaniya ot 1:
  do  1: 0
  do  2: 21
  do  3: 53
  do  4: 96
  do  5: 150
  do  6: 169
  do  7: 93
  do  8: 128
  do  9: 30
  do 10: 11
  do 11: 285
  do 12: 266
  do 13: 446
  do 14: 411
```

```
Krachaishie puti ot 1:
  do  2: 1 -> 2   (dlina 21)
  do  3: 1 -> 2 -> 3   (dlina 53)
  do  4: 1 -> 2 -> 3 -> 4   (dlina 96)
  do  5: 1 -> 2 -> 3 -> 4 -> 5   (dlina 150)
  do  6: 1 -> 2 -> 7 -> 6   (dlina 169)
  do  7: 1 -> 2 -> 7   (dlina 93)
  do  8: 1 -> 10 -> 9 -> 8   (dlina 128)
  do  9: 1 -> 10 -> 9   (dlina 30)
  do 10: 1 -> 10   (dlina 11)
  do 11: 1 -> 2 -> 7 -> 6 -> 11   (dlina 285)
  do 12: 1 -> 2 -> 3 -> 12   (dlina 266)
  do 13: 1 -> 10 -> 9 -> 8 -> 13   (dlina 446)
  do 14: 1 -> 14   (dlina 411)
```

```
Krachaishie rasstoyaniya ot 1 (Floyd):
  do  1: 0
  do  2: 21
  do  3: 53
  do  4: 96
  do  5: 150
  do  6: 169
  do  7: 93
  do  8: 128
  do  9: 30
  do 10: 11
  do 11: 285
  do 12: 266
  do 13: 446
  do 14: 411
```

```
=== ALGORITM FLOYDUWARSHALL ===
Matritsa krachaishih rasstoyaniy:
        1        2        3        4        5        6        7        8        9       10       11       12       13       14
 1:     0       21       53       96      150      169       93      128       30       11      285      266      446      411
 2:    21        0       32       75      129      148       72      149       51       32      264      245      467      432
 3:    53       32        0       43       97      162      104      181       83       64      278      213      499      464
 4:    96       75       43        0       54      119      147      192       94      107      235      256      510      469
 5:   150      129       97       54        0       65      141      228      148      161      181      310      546      415
 6:   169      148      162      119       65        0       76      163      199      180      116      375      481      480
 7:    93       72      104      147      141       76        0       87      123      104      192      317      405      504
 8:   128      149      181      192      228      163       87        0       98      117      279      394      318      539
 9:    30       51       83       94      148      199      123       98        0       19      315      296      416      441
10:    11       32       64      107      161      180      104      117       19        0      296      277      435      422
11:   285      264      278      235      181      116      192      279      315      296        0      491      597      596
12:   266      245      213      256      310      375      317      394      296      277      491        0      712      677
13:   446      467      499      510      546      481      405      318      416      435      597      712        0      857
14:   411      432      464      469      415      480      504      539      441      422      596      677      857        0
```

Тест15 – 1024vershin

```
Fayl: tests/test15.txt
Vershin: 1024, ryober: 1954

=== ALGORITM DEYKSTRY (ot vershiny 1) ===
Krachaishie rasstoyaniya ot 1:
  do  1: 0
  do  2: 3
  do  3: 8
  do  4: 15
  do  5: 24
  do  6: 35
  do  7: 48
  do  8: 63
  do  9: 80
  do 10: 99
```

```
12221          10184          8145
     42567          40560          38551
          10215          8176

Krachaishie rasstoyaniya ot 1 (Floyd):
  do  1: 0
  do  2: 3
  do  3: 8
  do  4: 15
  do  5: 24
  do  6: 35
  do  7: 48
  do  8: 63
  do  9: 80
  do 10: 99
  do 11: 120
  do 12: 143
  do 13: 168
```

```
do 289: 1 -> 33 -> 65 -> 97 -> 129 -> 161 -> 193 -> 225 -> 257 -> 289   (dlina 2610)
do 290: 1 -> 2 -> 34 -> 66 -> 98 -> 130 -> 162 -> 194 -> 226 -> 258 -> 290   (dlina 2631)
do 291: 1 -> 2 -> 3 -> 35 -> 67 -> 99 -> 131 -> 163 -> 195 -> 227 -> 259 -> 291   (dlina 2654)
do 292: 1 -> 2 -> 3 -> 4 -> 36 -> 68 -> 100 -> 132 -> 164 -> 196 -> 228 -> 260 -> 292   (dlina 2679)
do 293: 1 -> 2 -> 3 -> 4 -> 5 -> 37 -> 69 -> 101 -> 133 -> 165 -> 197 -> 229 -> 261 -> 293   (dlina 2706)
do 294: 1 -> 2 -> 3 -> 4 -> 5 -> 6 -> 38 -> 70 -> 102 -> 134 -> 166 -> 198 -> 230 -> 262 -> 294   (dlina 2735)
do 295: 1 -> 2 -> 3 -> 4 -> 5 -> 6 -> 7 -> 39 -> 71 -> 103 -> 135 -> 167 -> 199 -> 231 -> 263 -> 295   (dlina 2766)
do 296: 1 -> 2 -> 3 -> 4 -> 5 -> 6 -> 7 -> 8 -> 40 -> 72 -> 104 -> 136 -> 168 -> 200 -> 232 -> 264 -> 296   (dlina 2799)
do 297: 1 -> 2 -> 3 -> 4 -> 5 -> 6 -> 7 -> 8 -> 9 -> 41 -> 73 -> 105 -> 137 -> 169 -> 201 -> 233 -> 265 -> 297   (dlina 2834)
do 298: 1 -> 2 -> 3 -> 4 -> 5 -> 6 -> 7 -> 8 -> 9 -> 10 -> 42 -> 74 -> 106 -> 138 -> 170 -> 202 -> 234 -> 266 -> 298   (dlina 2871)
do 299: 1 -> 2 -> 3 -> 4 -> 5 -> 6 -> 7 -> 8 -> 9 -> 10 -> 11 -> 43 -> 75 -> 107 -> 139 -> 171 -> 203 -> 235 -> 267 -> 299   (dlina 2910)
do 300: 1 -> 2 -> 3 -> 4 -> 5 -> 6 -> 7 -> 8 -> 9 -> 10 -> 11 -> 12 -> 44 -> 76 -> 108 -> 140 -> 172 -> 204 -> 236 -> 268 -> 300   (dlina
```

Вывод: реализована программа на C для поиска кратчайших путей в взвешенном графе алгоритмами Дейкстры и Флойда–Уоршелла. Проверена на 15 тестах — от малых графов до графа с 1024 вершинами. Результаты совпали с теорией.